SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: OBJECT ORIENTED ANALYSIS AND DESIGN
COURSE CODE: CSA1106

COURSE OUTCOME COVERED

CO2: Design UML diagrams to represent system requirements and architecture. *(BL3)*

Assessment Tool Weightage: 40%

QUESTIONS: Any 4 **Total Marks:40**

Annexure A

ANALYTICAL PROBLEM SOLVING

Code of Conduct:

I D Raghul 192511115 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

ANALYTICAL PROBLEM SOLVING

1. Use Case Diagram Problem

A **Smart Banking System** allows customers to create accounts, transfer funds, pay bills, and view transaction history. Bank employees verify customer details, while administrators manage user accounts and banking services. The system requirements are only partially documented.

Tasks:
a) Identify all actors involved in the system.
b) List the primary use cases.
c) Define relationships (include, extend, generalization).
d) Construct a complete **Use Case Diagram** representing system functionality.

(10 Marks)

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2. Class Diagram Problem

An **Online Vehicle Rental System** consists of entities such as Vehicle, Customer, Reservation, Payment, and RentalAgent. The relationships among these entities have not been clearly defined.

Tasks:
a) Identify classes with appropriate attributes and methods.
b) Define relationships (association, aggregation, inheritance, composition where applicable).
c) Specify multiplicity between classes.
d) Design a detailed **Class Diagram** for the system.

(10 Marks)

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3. Sequence Diagram Problem

An **E-Commerce Shopping System** allows a customer to browse products, add items to the cart, place an order, make payment, and receive order confirmation. The sequence of interactions between system components is unclear.

Tasks:
a) Identify all objects participating in the interaction.
b) Define the message flow between objects.
c) Represent activation bars and lifelines.
d) Construct a **Sequence Diagram** showing the complete interaction flow.

(10 Marks)

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4. Activity Diagram Problem

A **Job Recruitment Portal** enables applicants to register, upload resumes, apply for jobs, attend interviews, and receive offer letters. Recruiters verify applications before scheduling interviews. The business workflow needs optimization.

Tasks:
a) Identify all activities in the recruitment workflow.
b) Add decision nodes for selection and rejection.
c) Include parallel activities wherever applicable.
d) Design an optimized **Activity Diagram** representing the complete workflow.

(10 Marks)

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5. State Diagram Problem

A **Smart Parking Management System** operates through states such as Parking Slot Available, Vehicle Entered, Parking Allocated, Payment Pending, Payment Completed, and Vehicle Exited. The system behavior during parking operations needs to be modeled.

Tasks:
a) Identify all possible states of the system.
b) Define events triggering state transitions.
c) Include initial and final states.
d) Construct a **State Transition Diagram** representing system behavior.

(10 Marks)

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6. Deployment Diagram Problem

A **Smart Agriculture Monitoring System** uses IoT sensors to monitor soil moisture, temperature, and humidity. Sensor data is transmitted to a cloud platform where farmers monitor crop conditions through a mobile application. Automated irrigation is activated whenever soil moisture falls below a predefined threshold.

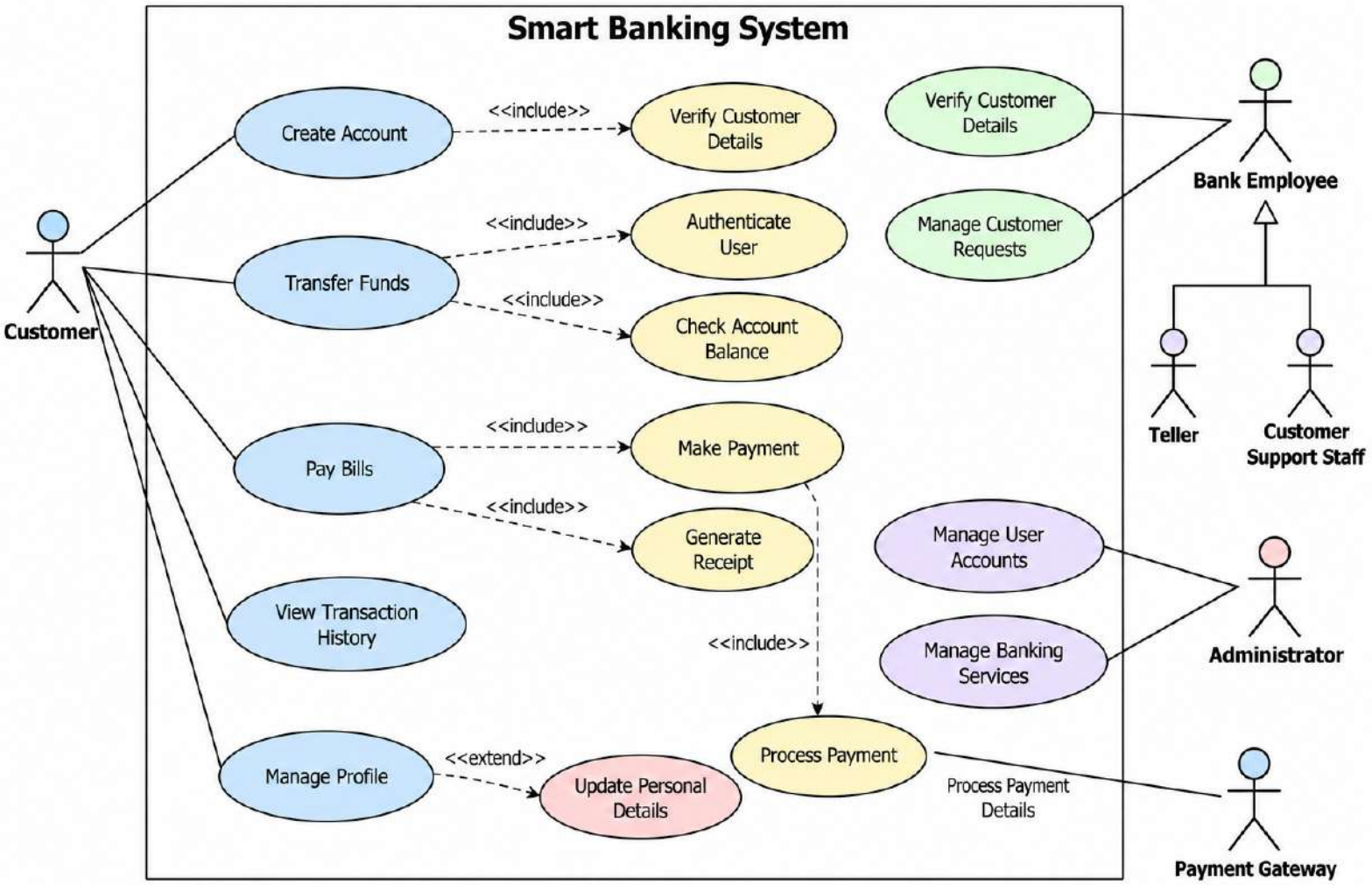
Tasks:
a) Identify all hardware nodes involved in the system.
b) Specify software components deployed on each node.
c) Define communication links between hardware nodes.
d) Construct a complete **Deployment Diagram** representing the system architecture.

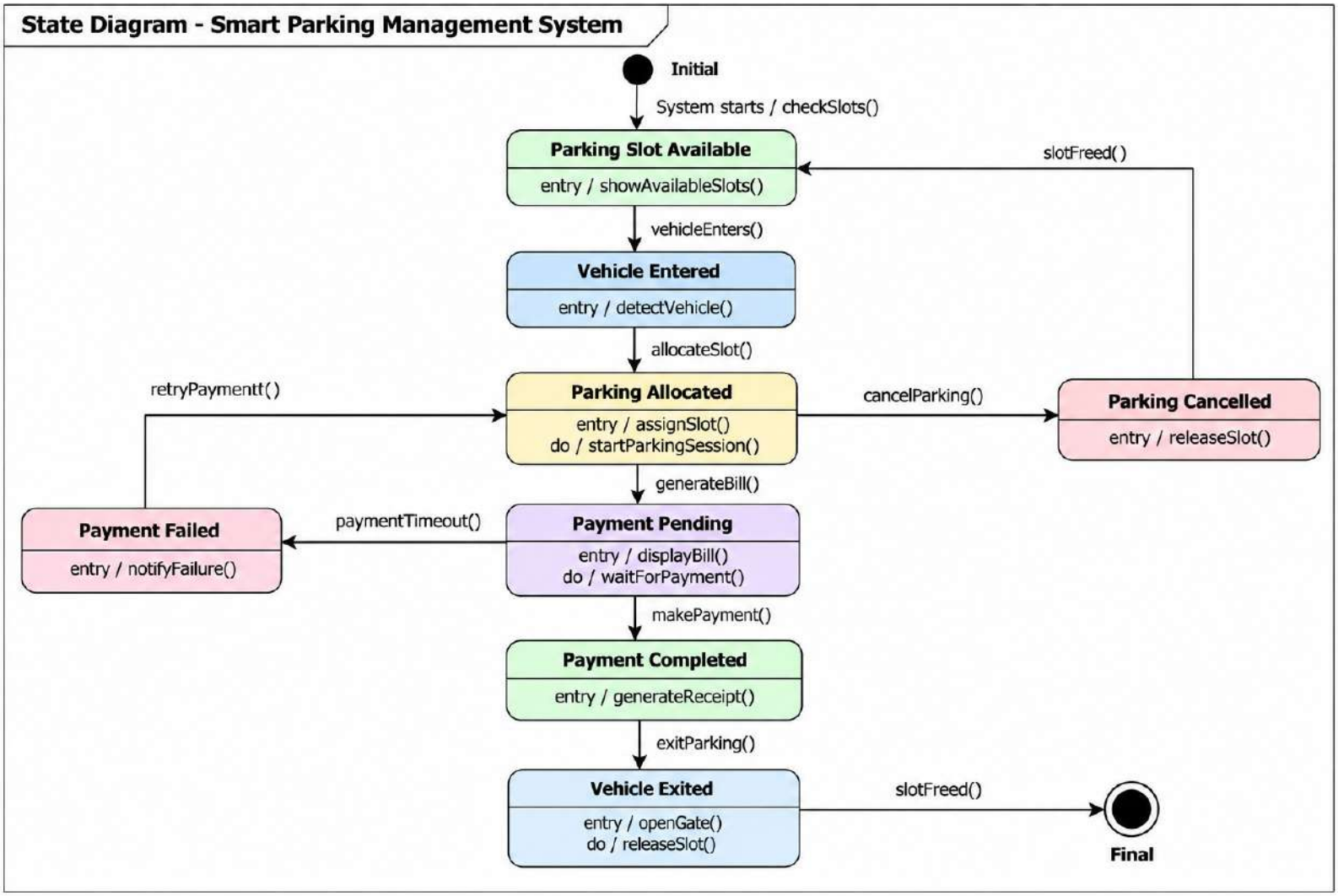
(10 Marks)

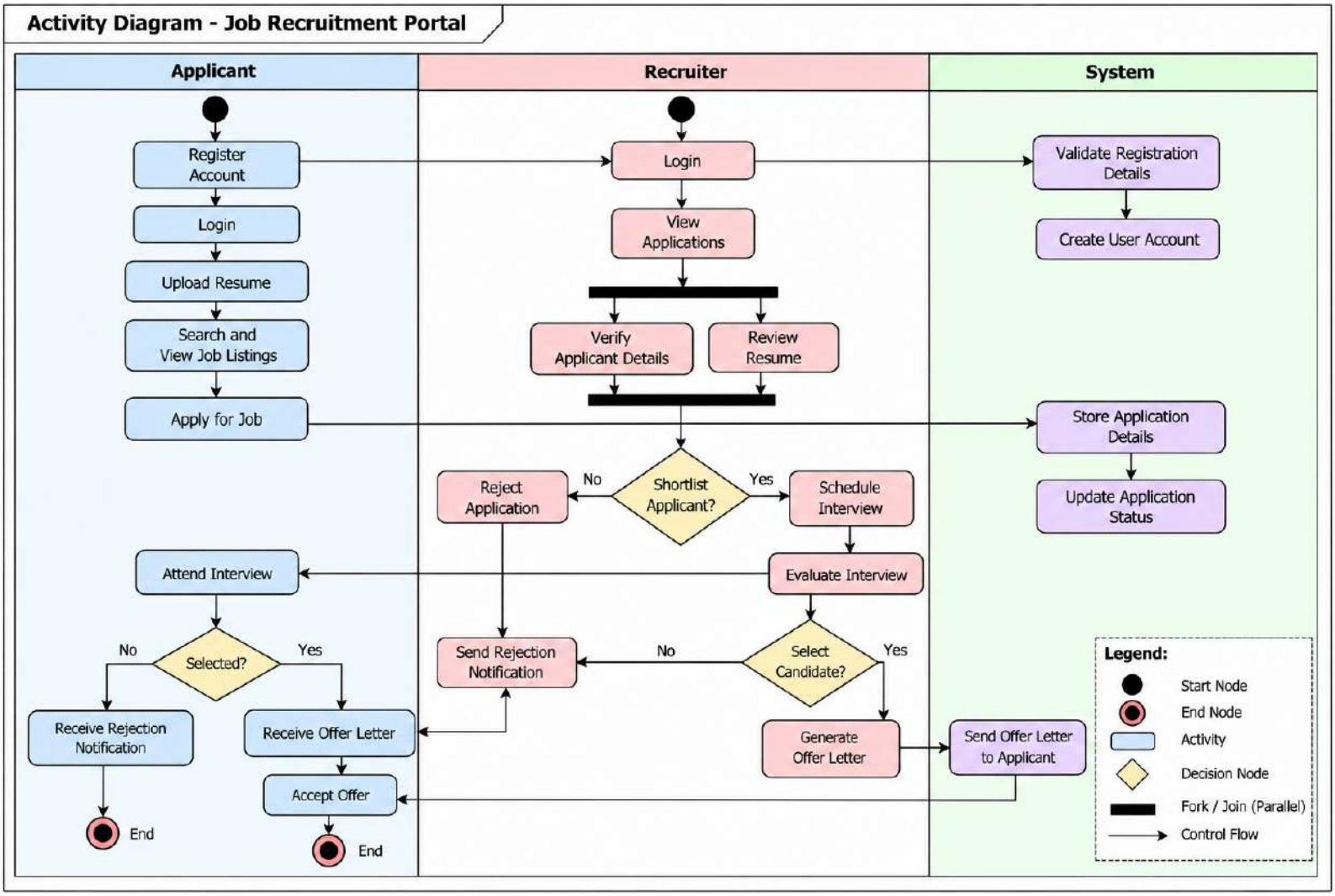
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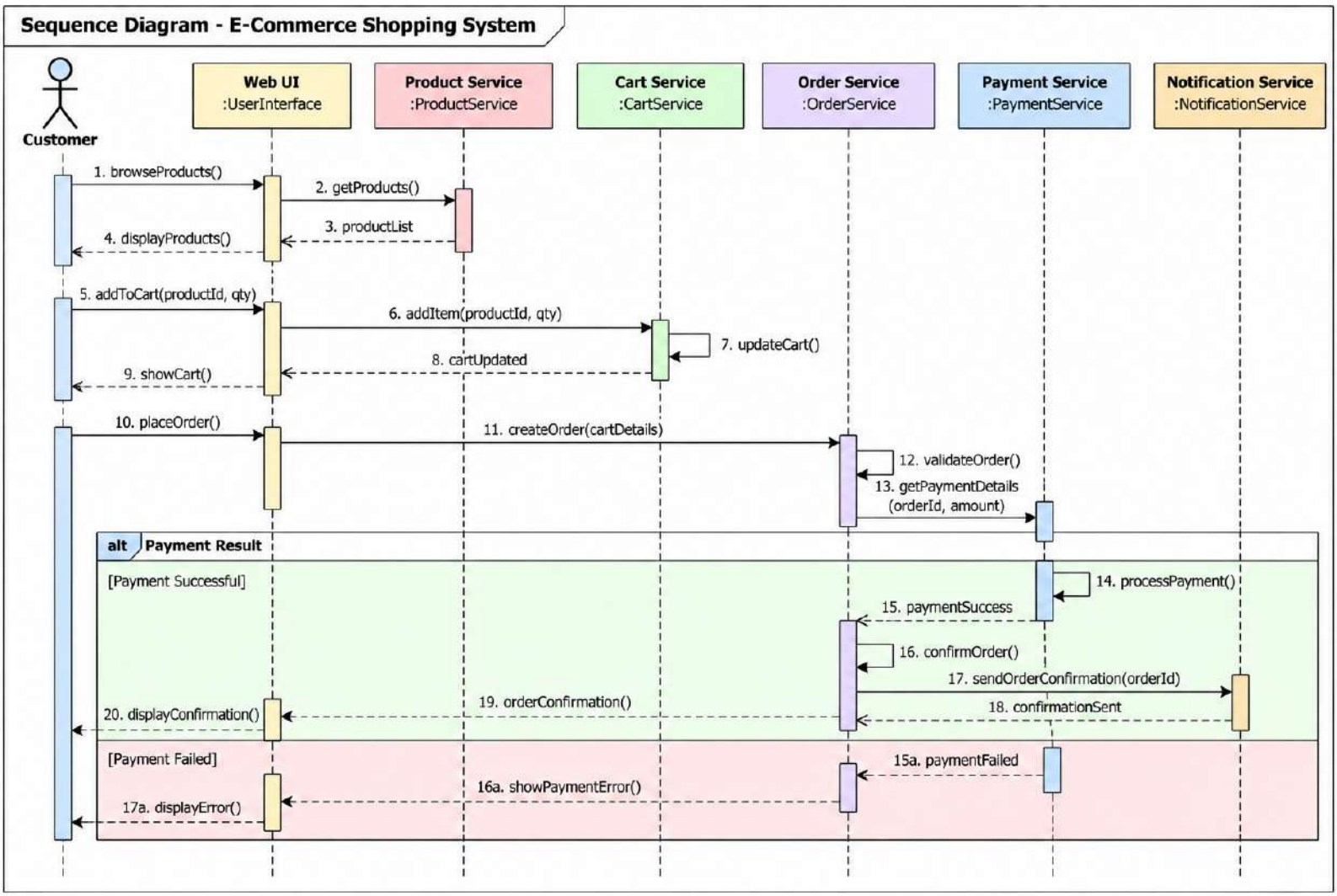
ANALYTICAL PROBLEM SOLVING

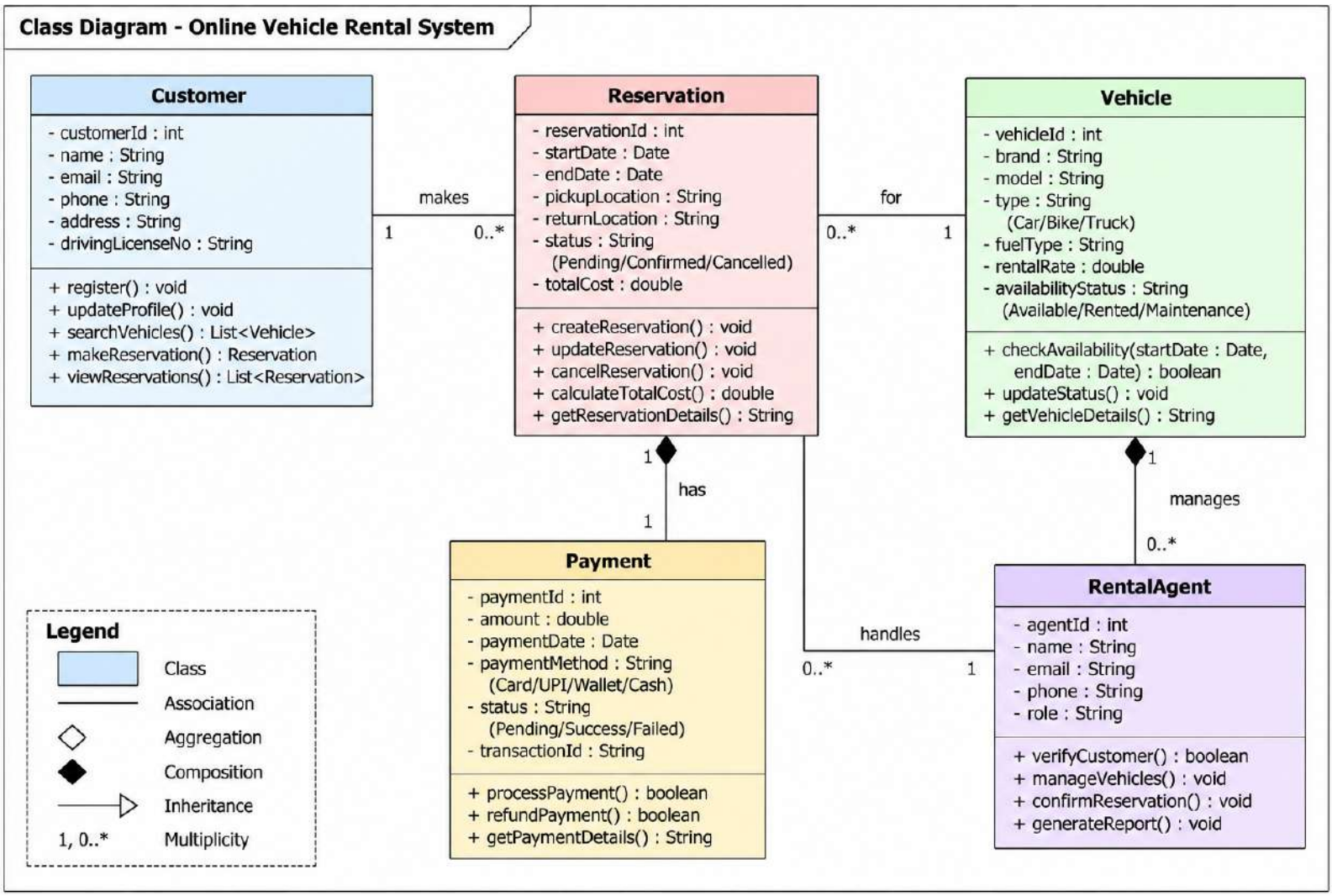
Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

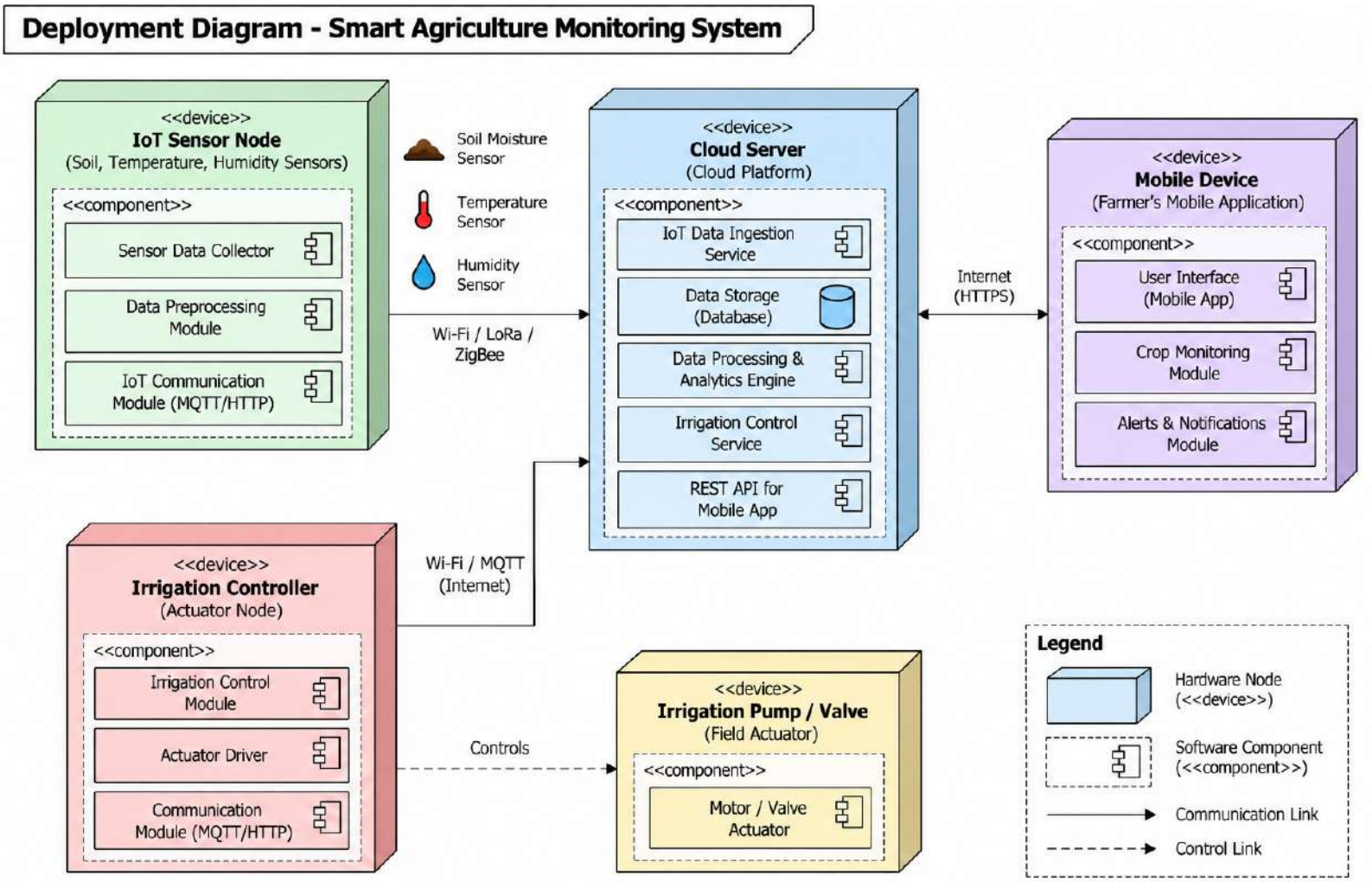

Smart Banking System
Create Account
Verify Customer Details
<<include>>
Transfer Funds
Authenticate User
<<include>>
Check Account Balance
<<include>>
Pay Bills
Make Payment
<<include>>
Generate Receipt
<<include>>
View Transaction History
Manage Profile
<<extend>>
Update Personal Details
Process Payment
Process Payment Details
Verify Customer Details
Manage Customer Requests
Manage User Accounts
Manage Banking Services
Customer
Bank Employee
Teller
Customer Support Staff
Administrator
Payment Gateway


State Diagram - Smart Parking Management System
Initial
System starts / checkSlots()
Parking Slot Available
entry / showAvailableSlots()
slotFreed()
vehicleEnters()
Vehicle Entered
entry / detectVehicle()
allocateSlot()
retryPaymentt()
Parking Allocated
entry / assignSlot()
do / startParkingSession()
cancelParking()
Parking Cancelled
entry / releaseSlot()
generateBill()
Payment Failed
entry / notifyFailure()
paymentTimeout()
Payment Pending
entry / displayBill()
do / waitForPayment()
makePayment()
Payment Completed
entry / generateReceipt()
exitParking()
Vehicle Exited
entry / openGate()
do / releaseSlot()
slotFreed()
Final


Activity Diagram - Job Recruitment Portal
Applicant
Recruiter
System
Register Account
Login
Upload Resume
Search and View Job Listings
Apply for Job
Attend Interview
Selected?
No
Yes
Receive Rejection Notification
End
Receive Offer Letter
Accept Offer
End
Login
View Applications
Verify Applicant Details
Review Resume
Shortlist Applicant?
No
Yes
Reject Application
Schedule Interview
Evaluate Interview
Select Candidate?
No
Yes
Send Rejection Notification
Generate Offer Letter
Validate Registration Details
Create User Account
Store Application Details
Update Application Status
Send Offer Letter to Applicant
Legend:
Start Node
End Node
Activity
Decision Node
Fork / Join (Parallel)
Control Flow


Sequence Diagram - E-Commerce Shopping System
Customer
Web UI
:UserInterface
Product Service
:ProductService
Cart Service
:CartService
Order Service
:OrderService
Payment Service
:PaymentService
Notification Service
:NotificationService
1. browseProducts()
2. getProducts()
3. productList
4. displayProducts()
5. addToCart(productId, qty)
6. addItem(productId, qty)
7. updateCart()
8. cartUpdated
9. showCart()
10. placeOrder()
11. createOrder(cartDetails)
12. validateOrder()
13. getPaymentDetails
(orderId, amount)
alt
Payment Result
[Payment Successful]
14. processPayment()
15. paymentSuccess
16. confirmOrder()
17. sendOrderConfirmation(orderId)
18. confirmationSent
19. orderConfirmation()
20. displayConfirmation()
[Payment Failed]
15a. paymentFailed
16a. showPaymentError()
17a. displayError()


Class Diagram - Online Vehicle Rental System
Customer
- customerId : int
- name : String
- email : String
- phone : String
- address : String
- drivingLicenseNo : String
+ register() : void
+ updateProfile() : void
+ searchVehicles() : List<Vehicle>
+ makeReservation() : Reservation
+ viewReservations() : List<Reservation>
makes
1
0..*
Reservation
- reservationId : int
- startDate : Date
- endDate : Date
- pickupLocation : String
- returnLocation : String
- status : String
(Pending/Confirmed/Cancelled)
- totalCost : double
+ createReservation() : void
+ updateReservation() : void
+ cancelReservation() : void
+ calculateTotalCost() : double
+ getReservationDetails() : String
for
0..*
1
Vehicle
- vehicleId : int
- brand : String
- model : String
- type : String
(Car/Bike/Truck)
- fuelType : String
- rentalRate : double
- availabilityStatus : String
(Available/Rented/Maintenance)
+ checkAvailability(startDate : Date,
endDate : Date) : boolean
+ updateStatus() : void
+ getVehicleDetails() : String
1
has
1
Payment
- paymentId : int
- amount : double
- paymentDate : Date
- paymentMethod : String
(Card/UPI/Wallet/Cash)
- status : String
(Pending/Success/Failed)
- transactionId : String
+ processPayment() : boolean
+ refundPayment() : boolean
+ getPaymentDetails() : String
handles
0..*
1
1
manages
0..*
RentalAgent
- agentId : int
- name : String
- email : String
- phone : String
- role : String
+ verifyCustomer() : boolean
+ manageVehicles() : void
+ confirmReservation() : void
+ generateReport() : void
Legend
Class
Association
Aggregation
Composition
Inheritance
1, 0..*
Multiplicity


Deployment Diagram - Smart Agriculture Monitoring System
<<device>>
IoT Sensor Node
(Soil, Temperature, Humidity Sensors)
<<component>>
Sensor Data Collector
Data Preprocessing Module
IoT Communication Module (MQTT/HTTP)
Soil Moisture Sensor
Temperature Sensor
Humidity Sensor
Wi-Fi / LoRa / ZigBee
<<device>>
Cloud Server
(Cloud Platform)
<<component>>
IoT Data Ingestion Service
Data Storage (Database)
Data Processing & Analytics Engine
Irrigation Control Service
REST API for Mobile App
Internet (HTTPS)
<<device>>
Mobile Device
(Farmer's Mobile Application)
<<component>>
User Interface (Mobile App)
Crop Monitoring Module
Alerts & Notifications Module
Wi-Fi / MQTT (Internet)
<<device>>
Irrigation Controller
(Actuator Node)
<<component>>
Irrigation Control Module
Actuator Driver
Communication Module (MQTT/HTTP)
Controls
<<device>>
Irrigation Pump / Valve
(Field Actuator)
<<component>>
Motor / Valve Actuator
Legend
Hardware Node (<<device>>)
Software Component (<<component>>)
Communication Link
Control Link